Computer Systems Modelling: Lent 2025 **Assessment 1**
Instructor: Prof. S. Keshav 10-11 AM, February 27, 2025
Answer all questions. Please return this sheet at the end of the assessment. Maximum marks 15

Question 1. You are asked to model the behaviour of a web surfer going from website to website as a stochastic process.

a) Is this process (i) discrete or continuous space (ii) discrete or continuous time? Justify your choices. **(1 mark)**

b) What assumption(s) do you need to make to make the process a homogenous Markov process. Give one example of a situation where each of these assumptions may not hold **(2 marks)**

c) Given your assumptions, you observe that the mean time spent at a website is 5 time units. What is the probability that the surfer spends 10 time units at a website? **(1 mark)**

Solution:
(a) This process is:

- **Discrete space** - The state space consists of distinct websites that can be visited, making it countable and discrete.
- **Continuous time** - The time spent on each website is a continuous variable that can take any positive real value.

Justification: While websites are distinct entities (you cannot be on 0.5 of a website), the duration of visits can be any real number (one can spend 2.37 minutes on a site).
(b) To make this a homogeneous Markov process, we need:

1. **Memoryless property:** The future state depends only on the current state, not on the history of previous states.
   Example violation: If the surfer uses browser history to avoid revisiting recent sites, the process depends on history.
2. **Time homogeneity:** Transition probabilities remain constant over time.
   Example violation: If the surfer's behavior changes between day and night (e.g., more likely to visit entertainment sites at night vs. work sites during day).
3. **Independence of time spent:** The time spent on a site doesn't affect transition probabilities.
   Example violation: If a long time spent on one site indicates high interest, making the user more likely to click related links.

(c) The probability of spending *exactly* 10 time units is 0. For the probability of spending *at least* 10 time units, since residence time in a continuous time Markov chain is exponentially distributed:

$$P(T > 10) = e^{t/\lambda} = e^{-\frac{10}{5}} = e^{-2} = 0.135$$

Question 2.
Consider a continuous-time Markov chain with states {0, 1, 2} and the following transition rate matrix:

$$Q = \begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

a) Calculate the stationary distribution $\pi = [\pi_0, \pi_1, \pi_2]$. **(1 mark)**

b) Calculate the expected time spent in state 0 before transitioning to another state. **(0.5 marks)**

Solution: **Part (a):** To find the stationary distribution π, we need to solve the system of equations $\pi Q = 0$ along with the normalization condition $\sum_{i=0}^{2} \pi_i = 1$.
From $\pi Q = 0$, we get three equations:

$$\begin{aligned} -3\pi_0 + \pi_1 + \pi_2 &= 0 \\ 2\pi_0 - 2\pi_1 + \pi_2 &= 0 \\ \pi_0 + \pi_1 - 2\pi_2 &= 0 \end{aligned}$$

Along with the normalization condition:

$$\pi_0 + \pi_1 + \pi_2 = 1$$

Let's solve this step by step:
From the first equation:

$$\pi_1 + \pi_2 = 3\pi_0 \tag{1}$$

From the second equation:

$$2\pi_0 - 2\pi_1 + \pi_2 = 0$$

Rearranging:

$$\pi_2 = 2\pi_1 - 2\pi_0 \tag{2}$$

Substituting (2) into (1):

$$\begin{aligned} \pi_1 + (2\pi_1 - 2\pi_0) &= 3\pi_0 \\ 3\pi_1 - 2\pi_0 &= 3\pi_0 \\ 3\pi_1 &= 5\pi_0 \\ \pi_1 &= \frac{5}{3}\pi_0 \end{aligned} \tag{3}$$

Substituting (3) back into (2):

$$\begin{aligned} \pi_2 &= 2(\frac{5}{3}\pi_0) - 2\pi_0 \\ \pi_2 &= \frac{10}{3}\pi_0 - 2\pi_0 \\ \pi_2 &= \frac{4}{3}\pi_0 \end{aligned} \tag{4}$$

Using the normalization condition:

$$\begin{aligned} \pi_0 + \frac{5}{3}\pi_0 + \frac{4}{3}\pi_0 &= 1 \\ \pi_0(1 + \frac{5}{3} + \frac{4}{3}) &= 1 \\ \pi_0(\frac{12}{3}) &= 1 \\ \pi_0 &= \frac{1}{4} \end{aligned}$$

Therefore:

$$\pi_0 = \frac{1}{4}$$
$$\pi_1 = \frac{5}{12}$$
$$\pi_2 = \frac{1}{3}$$

Part (b): The expected time spent in state 0 before transitioning to another state is the reciprocal of the total rate of leaving state 0. This rate is given by the negative of the diagonal entry in the first row of Q, which is 3.
Therefore:

$$E[T_0] = \frac{1}{3} \text{ time units}$$

This interpretation makes intuitive sense because the process leaves state 0 at rate 2 to state 1 and rate 1 to state 2, for a total rate of 3, making the expected waiting time $\frac{1}{3}$.

Question 3.
Consider a birth-death process modeling a population where births occur at rate $\lambda = 2$ per unit time and deaths occur at rate $\mu = 1$ per unit time, regardless of the population size.

a) Calculate the approximate probability that the population size increases by exactly 2 in a time interval of length 1. **(2 marks)**

b) What is the long-term average population size? **(0.5 marks)**

Solution: **(a)** To find the probability of exactly 2 net increases in population during 1 time unit, we need to consider all combinations of births and deaths that result in a net increase of 2.
Let B(t) denote births and D(t) denote deaths in time t. For a net increase of 2:

$$B(1) - D(1) = 2$$

The probability is the sum of all possible combinations:

$$P(\text{Net increase} = 2) = \sum_{k \geq 2} P(B(1) = k) \times P(D(1) = k - 2)$$

Using Poisson probabilities:

$$P(B(1) = k) = \frac{e^{-2} \cdot 2^k}{k!}$$

$$P(D(1) = k) = \frac{e^{-1} \cdot 1^k}{k!}$$

Calculating the first few significant terms:

$$k = 2 : P(B = 2)P(D = 0) = \frac{2^2 e^{-2}}{2!} \cdot e^{-1}$$
$$k = 3 : P(B = 3)P(D = 1) = \frac{2^3 e^{-2}}{3!} \cdot \frac{e^{-1}}{1!}$$
$$k = 4 : P(B = 4)P(D = 2) = \frac{2^4 e^{-2}}{4!} \cdot \frac{e^{-1}}{2!}$$

Adding these terms:

$$P(\text{Net increase} = 2) \approx 0.182$$

(b) For the long-term average population size:
Since $\lambda > \mu$ (2 ¿ 1), this is a transient birth-death process. The expected population size grows linearly with time:

$$E[X(t)] = X(0) + (\lambda - \mu)t \to \infty \text{ as } t \to \infty$$

Therefore, the long-term average population size does not exist, as the population grows without bound at an average rate of $\lambda - \mu = 1$ per time unit.

Question 4. Consider an M/M/1/K queueing system where $\lambda = 50$ customers/hour, $\mu = 60$ customers/hour, and $K = 10$ (including the customer being served).

a) Calculate the probability that the system is full, i.e., the blocking probability. **(1 mark)**

b) Derive an expression for the *effective* arrival rate λ_{eff}, considering that some customers are blocked when the system is full. **(1 mark)**

c) You are given that the mean number of customers in the system is 3.9748. Use this to calculate the average time a customer spends in the system (including service time) for those customers who are not blocked. **(1 mark)**

d) How would you modify the system to reduce the blocking probability to less than 1%? Justify your answer with calculations. **(1.5 marks)**

Solution: a) For an M/M/1/K queue, the blocking probability is given by:

$$P_K = \frac{\rho^K(1-\rho)}{1-\rho^{K+1}}$$

where $\rho = \frac{\lambda}{\mu} = \frac{50}{60} \approx 0.8333$

$$P_{10} = \frac{0.8333^{10}(1-0.8333)}{1-0.8333^{11}} \approx 0.031$$

The blocking probability is approximately 3.1%.
b) The effective arrival rate λ_{eff} is the rate at which customers are actually admitted to the system. It's given by:

$$\lambda_{eff} = \lambda(1 - P_K)$$

$$\lambda_{eff} = 50(1 - 0.031) = 48.445$$

customers/hour
c) The average time a customer spends in the system (for those not blocked) can be calculated using Little's Law:

$$W = \frac{L}{\lambda_{eff}}$$

$$W = \frac{3.9748}{48.445} \approx 0.82h$$

or 4.92 minutes

d) To reduce the blocking probability to less than 1%, we need to increase K. We can find the minimum K that satisfies:

$$\frac{\rho^K(1-\rho)}{1-\rho^{K+1}} < 0.01$$

Solving this numerically (by trying increasing values of K), we find that K = 17 gives a blocking probability of approximately 0.0098, which is just below 1%.
To justify: With K = 16,

$$P_{16} = \frac{0.8333^{16}(1-0.8333)}{1-0.8333^{17}} \approx 0.0098$$

Therefore, to reduce the blocking probability to less than 1%, the system should be modified to accommodate at least 16 customers (including the one being served).

Question 5.
Consider an M/G/1 queue where customers arrive according to a Poisson process with rate $\lambda = 10$ per hour. The service time S has the following distribution:

$$P(S=1) = 0.2, \quad P(S=2) = 0.3, \quad P(S=5) = 0.5 \quad \text{(in minutes)}$$

a) Calculate the mean and variance of the service time. **(1 mark)**

b) Using the Pollaczek-Khinchin formula, calculate the average number of customers in the queue.**(1 mark)**

c) Calculate the probability that an arriving customer finds the system empty.**(0.5 marks)**

Solution:

a) $E[S] = 0.2(1) + 0.3(2) + 0.5(5) = 3.3$ minutes $= 0.055$ hours
$E[S^2] = 0.2(1^2) + 0.3(2^2) + 0.5(5^2) = 13.9$ minutes^2
$Var(S) = E[S^2] - (E[S])^2 = 13.9 - 3.3^2 = 3.01$ minutes^2

b) $\rho = \lambda E[S] = 10(0.055) = 0.55$ $N = \frac{\lambda^2 \cdot E[S^2]}{2(1-\rho)} = \frac{100 \cdot 14.7/3600}{2(0.45)} \approx 1.13$ customers

c) $P(\text{system empty}) = 1 - \rho = 1 - 0.55 = 0.45$